Reconstructing Predator-Driven Extirpation Dynamics from Response-Only Signals

Synthetic benchmark and constrained mink→vole case study

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0.1 Abstract

We developed and validated a process-explicit, pattern-oriented framework for reconstructing prey extirpation timing from response-only data under sparse and biased observation. A synthetic mink→vole twin was used to separate ideal algorithmic identifiability (`TX_true`) from realistic inference after an observation model and first-passage TX estimation (`TX_obs`). We then applied the constrained model to the real mink–water-vole case.

The synthetic benchmark demonstrates where recovery is robust and where it breaks under low-information regimes. Stress sweeps expose failure regimes driven primarily by arrival-surface incompleteness and summary compatibility collapse. In the real-data case study, information audits and representation diagnostics quantify whether inference is supported at the chosen analysis unit.

All figures and tables are pinned to canonical artefact bundles, and the full workflow is reproducible via `targets`, `renv`, and Quarto.

1 Introduction

This manuscript targets a recurring applied challenge: reconstructing predator front dynamics and prey extirpation timing when direct predator abundance data are sparse, noisy, or unavailable. We use a response-only strategy based on prey extirpation timing (`TX`) derived from presence-only records and calibrate process parameters against pattern-oriented summaries.

The key scientific risk is confounding ecological signal with observation artifacts. To address this, we require a locked synthetic benchmark before any real-data claim. The benchmark quantifies what is recoverable under known truth and explicitly maps weak-identifiability and failure regimes.

The current focal system is invasive American mink (*Neovison vison*) and water vole (*Arvicola amphibius*) on a British National Grid representation. We treat this as both an ecological case and a methodological stress-test for response-only inference.

2 Methods

2.1 Process model and extirpation timing

The simulation core is a streaming TX-only engine that avoids storing full `N[x,y,t]` abundance cubes for large grids. Extirpation time is defined by a first-passage rule, and this definition is shared between synthetic generation and inference summaries.

Predator arrival is represented as a year surface `T_P(x)`. Predation intensity ramps after arrival and combines with local growth and dispersal dynamics to produce extirpation timing `TX(x)`.

2.2 Observation layer and TX estimator

Synthetic realistic mode converts latent dynamics into observation-limited signals and reconstructs `TX_obs` using the same first-passage estimator used in real data. This creates a strict separation between:

- ideal identifiability (`TX_true`),
- realistic identifiability (`TX_obs`).

2.3 Summary-based calibration (screen + ABC)

Inference uses response-oriented summary vectors describing coverage, lag, geometry, and coverage curves. A screen stage filters parameter draws by loss and compatibility criteria, then rejection ABC refines around shortlist regions.

2.4 Quality gates and canonical discipline

Every major run is validated with explicit gates and packaged in canonical artefact folders containing:

- `manifest.yaml` (schema version, commit SHA, config snapshots, seeds),
- rendered reports and primary tables/figures,
- session information.

These gates are treated as scientific safeguards, not only engineering checks.

3 Results

3.1 Synthetic benchmark lock-down

The synthetic benchmark was completed in two canonical stages:

- pilot-grid benchmark (`WP-0.3.5a`),
- GB-grid benchmark (`WP-0.3.5b`, graph mode).

Both runs include ideal and realistic modes with replicated recovery metrics and posterior predictive checks.

The GB-scale canonical run (`WP-0.3.5b`) passed all configured benchmark gates across replicates (`G1`–`G5`). In realistic mode, predictive summaries remained above gate thresholds, while core front parameters (`alpha`, `disp_mean_km`) were retained as weakly identifiable under the declared benchmark stance.

3.2 Stress regimes and identifiability limits

`WP-0.3.6` adds a compact stress grid over arrival realism, observation intensity, and predation strength multipliers. The resulting heatmap and gate tables identify where recovery collapses, with failures concentrated in arrival coverage and screen/ABC compatibility gates.

The stress sweep showed a sharp regime split: scenarios using `arrival_source = real` failed all-gate criteria (pass rate 0), while `arrival_source = synthetic_plane` with moderate observation intensity remained identifiable. This provides an explicit "information-limited failure map" for interpreting real-data outcomes.

3.3 Synthetic interpretation stance

The synthetic package is used as an identifiability map: it distinguishes recoverable from weakly identified parameters and sets expectations for real-data transfer under sparse arrival coverage.

3.4 Real-data information audit and representation triage

The real-data workflow begins with an information audit (`WP-0.4.0`) and representation audit (`WP-0.4.1`) before full constrained inference. This prevents over-interpreting runs with inadequate overlap between reconstructed `TX_obs` and finite predator-arrival support.

For the selected representation (`grid_all`), the audit indicates:

- `n_units = 3621`
- `n_TX_obs = 569` (`coverage_obs  0.157`)
- `n_TP_finite = 7`
- `n_pairs = 1` (`TX_obs` and `T_P` jointly finite)

So the unit-count gate passes, but arrival/response overlap is extremely sparse.

3.5 Constrained inference and ablation

The constrained real run (`WP-0.4.3`) combines screen, ABC, and posterior predictive checks, with ablation diagnostics (`WP-0.4.2`) used to decide whether optional random-effects structure materially improves fit.

The ablation showed no practical gain from adding `pred_scale` random effects: both `delta_only` and `delta_plus_re` produced zero compatible draws at the tested budget, with near-identical median loss and coverage.

3.6 Case-study verdict

Gate outcomes for the constrained run were uniformly negative (`best_r2 = NA`, `best_lag = NA`, `best_coverage  2.8e-4`, `abc_accept_rate = 0`), and no ABC posterior was retained under the configured thresholds.

The corresponding verdict is therefore **data-limited/mechanism-incomplete at this representation**, not a positive confirmation of predation-only sufficiency. This negative result is still informative because it is grounded in explicit audits, gates, and canonical artefacts rather than post hoc rationalisation.

4 Discussion

The central outcome of this workflow is not merely a fitted model; it is a transparent map of where inference is and is not supported.

Three practical lessons emerge:

- performance tractability must be engineered explicitly for large spatial domains;
- identifiability must be demonstrated under synthetic truth before ecological claims on real data;
- low-information real-data outcomes are scientifically meaningful when failure modes are diagnosed and documented.

Future extensions can add habitat covariates and alternative analysis-unit representations, but only after preserving the same canonical and gate-driven discipline.

5 Reproducibility

5.1 Canonical artefact policy

Primary manuscript claims must link to canonical bundles under `outputs/canonical/`, with run manifests carrying commit SHA, configuration snapshots, and seeds.

For this draft, synthetic claims are pinned to:

- `outputs/canonical/0.3/WP-0.3.5b/syn_20260209_134622_035b_full_r20/`
- `outputs/canonical/0.3/WP-0.3.6/syn_20260209_154821_0099/`

Real-case claims are pinned to:

- `outputs/canonical/0.4/WP-0.4.3/real_20260209_171134_7350/`

5.2 Build commands

- Synthetic benchmark/report: `make synthetic`
- Real pilot/report: `make pilot`
- Test suite: `make test`
- Manuscript render: `quarto render manuscript/mink_vole_manuscript.qmd`

5.3 Figure and table manifest

See `manuscript/figure_table_manifest.csv` for explicit pointers from manuscript items to canonical artefacts.

5.4 Freeze checklist

Release-freeze and handoff checks are tracked in `manuscript/reproducibility_freeze.md`.